

Roshan

Educational Platform for Balochistan

1. INTRODUCTION & PROBLEM STATEMENT

1.1 Background

Balochistan, the largest province of Pakistan, faces significant challenges in its educational sector. Despite various government initiatives, the quality of education remains a pressing concern. Students in Balochistan often grow up in environments where:

- Memorization is prioritized over understanding - The education system rewards rote learning rather than critical thinking
- Cheating is normalized - Due to pressure and lack of resources, many students resort to dishonest practices
- Limited access to quality resources - Many schools lack libraries, laboratories, and digital resources
- Cultural barriers - Traditional mindsets sometimes discourage curiosity and questioning

1.2 The Core Problem

The fundamental problem is not a lack of intelligence or capability among Balochistan's students, but rather:

"An educational culture that rewards memorization over understanding, and grades over genuine learning."

This culture leads to:

- Students who can recite information without comprehending it
- Graduates who lack critical thinking skills
- A workforce that is not equipped for the modern economy
- Perpetuation of poverty and underdevelopment

1.3 Target Audience

The Roshan platform is designed for:

Audience	Needs	How Roshan Helps
Students	Engaging, quality learning materials	Interactive lessons, videos, quizzes

Teachers	Resources to improve teaching	Content they can use and adapt
Scholars	Platform to share knowledge	Ability to contribute content
Parents	Quality education for children	Free, accessible learning platform

1.4 Why "Roshan"?

The name Roshan (روشن) means "light" or "enlightenment" in Urdu. The platform aims to:

- Illuminate minds with knowledge
- Enlighten students about the joy of learning
- Bring light to the educational darkness in Balochistan

2: Proposed Solution & Implementation

Roshan is a fully functional web-based platform built using PHP, MySQL, Bootstrap, and JavaScript. It features a clean, responsive design that works seamlessly on both desktop and mobile devices, ensuring that students across Balochistan can access it regardless of their device. The platform offers a rich set of features including multimedia lessons, interactive quizzes, video integration, and book recommendations. Each lesson is designed to spark curiosity and encourage critical thinking, with summary sections, key takeaways, and thought-provoking questions that guide students toward deeper understanding.

One of the core strengths of Roshan is its secure and intuitive Admin Panel, which allows administrators to manage all content efficiently. Through the admin interface, they can create, update, and delete lessons, books, categories, and products with ease. The platform also includes a shop section where students can purchase study guides and lesson bundles, adding a practical e-commerce dimension to the project. Additionally, Roshan offers user registration and login functionality, enabling students to track their learning progress, save favorite lessons, and engage with quizzes to test their understanding. The combination of rich content, interactivity, and user management makes Roshan a complete and scalable educational solution.

Behind the scenes, the platform is powered by a robust MySQL database with eleven interconnected tables that manage everything from user accounts and lesson content to orders and product inventory. The system is designed with security in mind,

featuring password hashing, session management, and input sanitization to protect user data. During development, AI tools like ChatGPT and GitHub Copilot were used to assist with content generation, code optimization, and debugging, which accelerated the development process and ensured high-quality outcomes. The result is a platform that is not only functional but also reliable, secure, and ready to support the educational needs of students and teachers alike.

3: Challenges, AI Tools & Conclusion

Throughout the development of Roshan, several challenges were encountered and successfully overcome. One of the early issues was related to file paths and image uploads, where images stored in the project root were not displaying correctly because the system was looking for them inside the public folder. This was resolved by adjusting the image URL logic and ensuring consistent path handling across the platform. Another significant challenge was the admin login system, which initially failed due to password hash mismatches. After debugging and resetting the password hash, the issue was fixed, and the admin panel became fully functional. Additionally, session management conflicts arose when the shop's cart functions attempted to start a session that was already active. This was resolved by adding proper session status checks before calling `session_start()`. These experiences taught valuable lessons in debugging, error handling, and system integration.

AI tools played a key role throughout the development lifecycle. ChatGPT was used extensively for generating lesson content, writing product descriptions, and brainstorming feature ideas. GitHub Copilot assisted with code completion and helped identify syntax errors and optimization opportunities. These tools significantly improved productivity and allowed me to focus on higher-level design and functionality. Without them, building a feature-rich platform like Roshan would have taken considerably more time and effort. Their use is documented in the report to demonstrate the practical application of AI in modern web development.

4: Conclusion

In conclusion, Roshan successfully addresses the educational challenges faced by students in Balochistan by providing an engaging, interactive, and accessible learning platform. It promotes genuine understanding over rote memorization and encourages students to think critically and independently. With its comprehensive features, secure admin panel, and scalable architecture, Roshan has the potential to make a lasting impact on the educational culture of the region. This project has not only strengthened my technical skills but also deepened my commitment to using technology for social

good. I am proud of what has been achieved and hopeful that Roshan will contribute to a brighter, more enlightened future for the students of Balochistan.